

final

Set Up

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.2.1      v readr      2.2.0
v forcats    1.0.1      v stringr    1.6.0
v ggplot2    4.0.3      v tibble     3.3.1
v lubridate  1.9.5      v tidyr      1.3.2
v purrr      1.2.2
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
library(DHARMA)
```

This is DHARMA 0.5.0. For overview type '?DHARMA'. For recent changes, type `news(package = 'DHARMA')`.

Note that the default setting in `simulateResiduals` was changed to conditional simulations since version 0.5.0. For more details, see `?simulateResiduals`.

```
library(ggeffects)
```

```
# creating new dataset
nests <- read.csv("../data/nest_data_final.csv")
```

Problem 1

a.

In part 1 the test they used a logistic regression test.

In part 2 the test used was a Chi square test of independence.

b.

Part 1: American Avocet microhabitat use was not determined by the distance of the microhabitat from the water.

Part 2: The American Avocet, Forster's Tern, and Black-necked Stilts all occupy different habitats within the managed pond, non-tidal marsh, salt pond, tidal marsh, or tidal mudflat study areas.

c.

One other variable that could influence the birds' habitat use is how deep the water is in different microhabitats. This is a continuous quantitative variable that would be measured in meters. It could be measured with a meter stick at the different sites. This could effect the American Avocet habitat use because the depth of the water could determine what prey live in the habitat and the resources available to the birds.

Problem 2

a.

```
# creating new object
nests_clean <- nests |>
  # filtering to only include AB, RRB, and BBR
  filter(ant_species %in% c("AB", "RRB", "BBR")) |>
  # Only using the 3 species
  mutate(
    ant_species = factor(
      ant_species,
      levels = c("AB",
                  "RRB",
                  "BBR")
    )
  )
```

```

),
# creating new column of scientific names
scientific_name = case_when(
  # adding in the scientific names for the species
  ant_species == "AB" ~ "C. sjostedti",
  ant_species == "RRB" ~ "C. mimosae",
  ant_species == "BBR" ~ "C. nigriceps"
)) |>
# selecting columns of interest
select(case_control, height_cm,
        scientific_name)
# showing structure of dataframe
str(nests_clean)

```

```

'data.frame': 270 obs. of 3 variables:
 $ case_control : int 1 0 0 0 0 1 0 0 0 1 ...
 $ height_cm : num 392 310 513 154 324 ...
 $ scientific_name: chr "C. mimosae" "C. mimosae" "C. mimosae" "C. mimosae" ...

```

```

# displaying 10 rows
slice_sample(nests_clean, n = 10)

```

	case_control	height_cm	scientific_name
1	0	236.0	C. mimosae
2	0	223.5	C. mimosae
3	1	181.5	C. mimosae
4	0	267.0	C. sjostedti
5	1	511.5	C. mimosae
6	1	396.0	C. mimosae
7	0	132.0	C. nigriceps
8	0	118.0	C. mimosae
9	0	170.5	C. mimosae
10	0	119.0	C. mimosae

b.

The 1s show when a tree had a nest in it and the 0 shows where there was an available tree with no nest.

c.

Tree height is a continuous quantitative variable measured in cm. It could affect the probability of nest occurrence because as the tree height increases so does the probability of there being a nest also increases because it becomes more and more removed from predators. I found this in the papers introduction.

Ant species is a categorical variable of either *C. sjostedi*, *C. mimosae*, or *C. nigriceps*. The species of ants might effect the probability of a nest because they might have different ways of defending themselves from the birds that drives away the birds from making a nest. I found this in the end of the introduction.

d.

model number	description	predictors	interaction term?
1	case_control ~ height_cm + ant_species	tree height and ant species	no
2	case_control ~ height_cm * ant_species	tree height and ant species and the interaction	yes

e.

```
# Model 1: no interaction
model1 <- glm(
  case_control ~ height_cm + scientific_name, # setting variables
  data = nests_clean, # assigning dataset
  family = binomial
)

# Model 2: with interaction
model2 <- glm(
  case_control ~ height_cm * scientific_name, # setting variables
  data = nests_clean, # assigning dataset
  family = binomial
)
```

f.

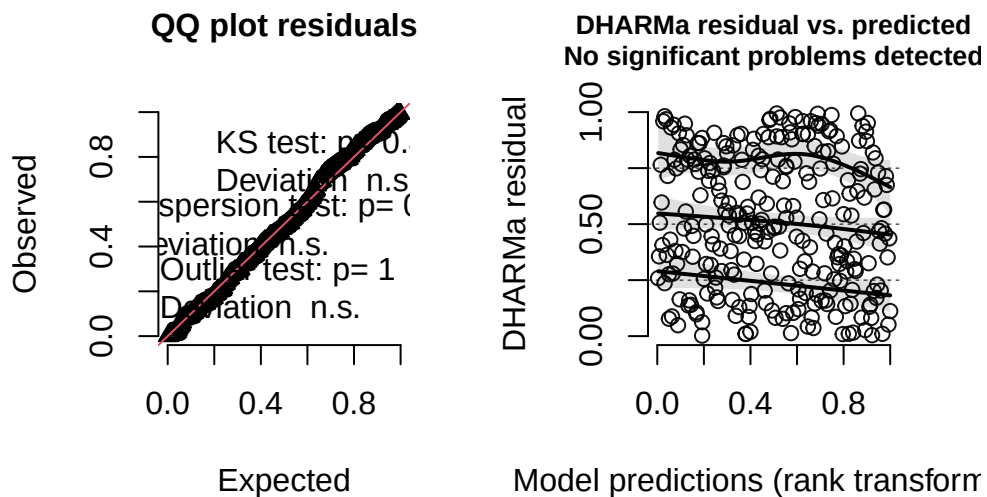
The better model is the interaction model with height times ant species because it had a lower AIC.

g.

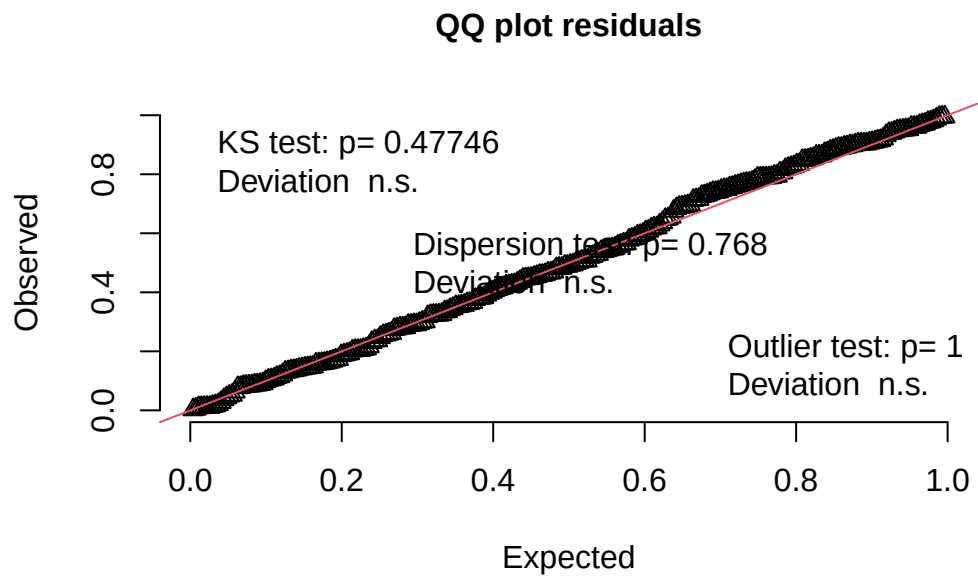
```
# simulate residuals from model2
model2_res <- simulateResiduals(fittedModel = model2)

# plotting the residuals
plot(model2_res)
```

DHARMa residual



```
# running diagnostic for uniformity
testUniformity(model2_res)
```

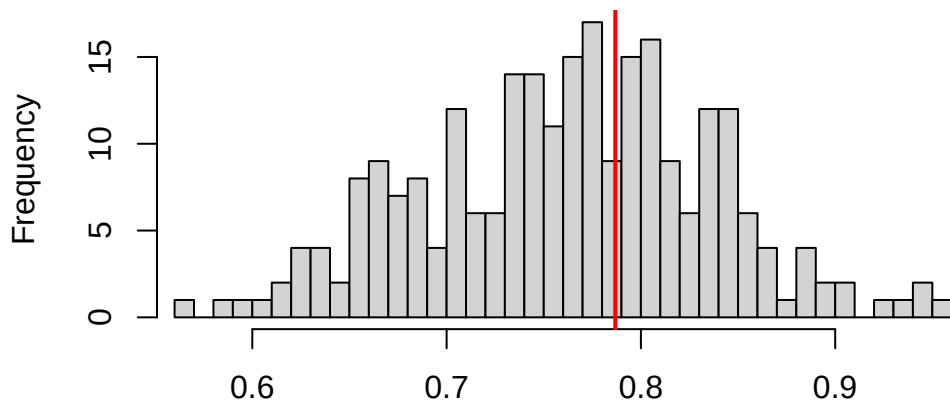


Asymptotic one-sample Kolmogorov-Smirnov test

```
data: simulationOutput$scaledResiduals  
D = 0.051246, p-value = 0.4775  
alternative hypothesis: two-sided
```

```
# running diagnostic for dispersing  
testDispersion(model2_res)
```

**DHARMA nonparametric dispersion test via sd of
residuals fitted vs. simulated**

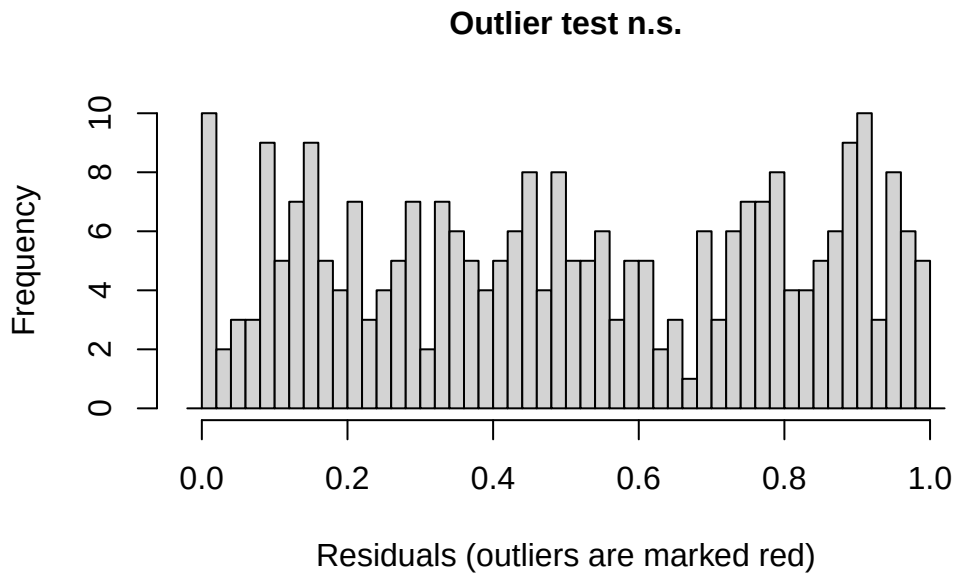


Simulated values, red line = fitted model. p-value (two.sided) = 0.768

DHARMA nonparametric dispersion test via sd of residuals fitted vs.
simulated

```
data: simulationOutput  
dispersion = 1.0325, p-value = 0.768  
alternative hypothesis: two.sided
```

```
# running diagnostic for outliers  
testOutliers(model2_res)
```



DHARMa bootstrapped outlier test

```
data: model2_res
outliers at both margin(s) = 0, observations = 270, p-value = 1
alternative hypothesis: two.sided
percent confidence interval:
 0 0
sample estimates:
outlier frequency (expected: 7.40740740740739e-05 )
0
```

h.

```
# creating predictions object from model 2
predictions <- ggpredict(
  model2,
  terms = c("height_cm", "scientific_name")
)
```

Some of the focal terms are of type `character`. This may lead to

unexpected results. It is recommended to convert these variables to factors before fitting the model.

The following variables are of type character: `scientific\_name`

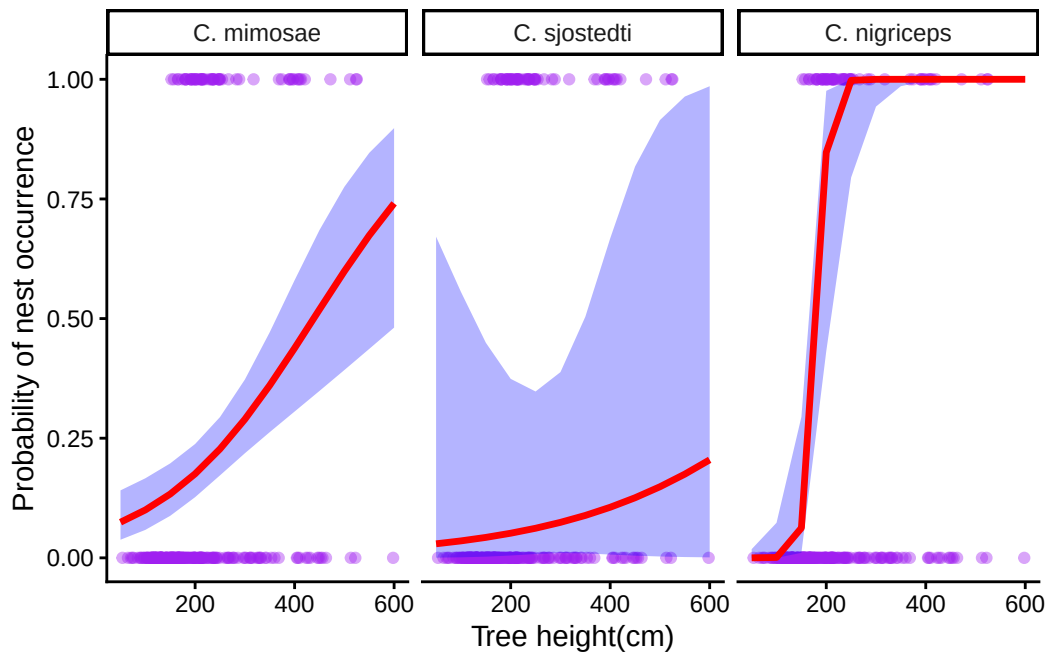
Data were 'prettified'. Consider using `terms="height\_cm [all]"` to get smooth plots.

```
# base layer: ggplot
ggplot() +
  # first layer: jitter
  geom_jitter(
    data = nests_clean,
    aes(
      x = height_cm,
      y = case_control
    ),
    # setting customizations
    width = 0,
    height = 0,
    color = "purple",
    alpha = 0.4
  ) +
  # second layer: ribbon
  geom_ribbon(
    data = predictions,
    aes(
      x = x,
      ymin = conf.low, # setting CI
      ymax = conf.high
    ),
    fill = "blue",
    alpha = 0.3
  ) +
  # third layer: line
  geom_line(
    data = predictions,
    aes(
      x = x,
      y = predicted
    ),
    color = "red",
    linewidth = 1.2
  )
```

```

) +
# seperating by ant species
facet_wrap(~group) +
# labeling x and y axis
labs(
  x = "Tree height(cm)",
  y = "Probability of nest occurrence"
) +
# retheming
theme_classic() +
  # removing legend
  theme(
    legend.position = "none"
  )

```



i.

Predicted probability of a bird nest being in a tree depending on the height and species of acacia ant. Purple points represent the data points of whether or not there was a nest in the tree. Red line shows the predicted probability at each tree height. The blue ribbon shows the 95% confidence interval. The first panel shows data for the *C. mimosae* species, the middle for the *C. sjostedti* species, and the third one for *C. nigriceps*. Each shows an increase in probability as

height increases with *C. nigriceps* being the steepest curve, then *C. mimosae*, then *C. sjostedti*. The data is from Palmer et al (2023).

j.

```
# new object
pred_600 <- data.frame(
  height_cm = 600, # setting predictor
  scientific_name = unique(nests_clean$scientific_name)
)
# predicting the probabilities
predict(
  model2,
  newdata = pred_600,
  type = "response")
```

	1	2	3
	0.7407857	0.2048017	1.0000000

k.